

Lecture 1: Choice Theory & Consumer Theory

Siguang Li

HKUST(GZ)

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The Consumer Problem

- Consumer chooses bundle $x = (x_1, \dots, x_n) \in \mathbb{R}_+^n$
- Problem:

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t.} \quad p \cdot x \leq w$$

- $p = (p_1, \dots, p_n)$ are prices
- w is wealth

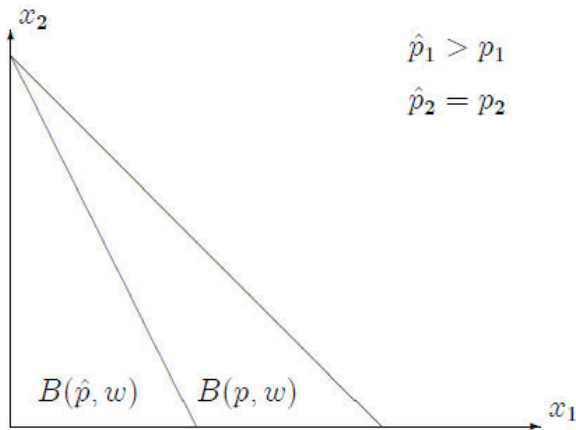
Budget Constraint

- Budget set:

$$B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\}$$

- Represents all affordable bundles
- Boundary: $p \cdot x = w$

Budget Set Illustration



- Two goods x_1, x_2
- Change in price shifts budget line

Observations on the Model

- Perfect information assumed
- Consumers are price-takers
- Prices are linear (no discounts)
- Goods are divisible
- Model can extend to indivisible goods

Marshallian Demand

- Solution $x(p, w)$ to consumer problem
- Also Walrasian / uncompensated demand
- In general: a set (correspondence), not always unique

Proposition 1: Budget Sets

Statement

- $B(\lambda p, \lambda w) = B(p, w)$ for all $\lambda > 0$
- If $p \gg 0$, then $B(p, w)$ compact

Proof (concise)

- Scaling cancels λ
- Closed and bounded set $\Rightarrow$ compact

Proposition 2: Existence

Statement

- If u continuous, $p \gg 0$, then solution exists

Proof

- $B(p, w)$ compact
- Continuous u attains maximum

Proposition 3: Homogeneity

Statement

- $x(\lambda p, \lambda w) = x(p, w)$
- Demand homogeneous of degree zero

Proof

- $B(\lambda p, \lambda w) = B(p, w)$
- Same feasible set $\Rightarrow$ same solution

Proposition 4: Walras' Law

Statement

- If preferences locally non-satiated
- Then for $x \in x(p, w)$: $p \cdot x = w$

Proof

- Suppose $p \cdot x < w$
- Local non-satiation $\Rightarrow$ exists $y \succ x$ with $p \cdot y < w$
- Contradiction

Proposition 5: Convexity and Uniqueness

Statement

- Convex preferences $\Rightarrow x(p, w)$ convex-valued
- Strictly convex $\Rightarrow$ unique solution

Proof

- If $x, x' \in x(p, w)$, convex combination feasible
- By convexity: also optimal
- Strict convexity $\Rightarrow$ only one optimal point

Proposition 6: Derivatives of Demand

Assume differentiable demand, preferences locally non-satiated
Results

- (1) Homogeneity condition:

$$\sum_j p_j \frac{\partial x_i}{\partial p_j} + w \frac{\partial x_i}{\partial w} = 0$$

- (2) Price changes:

$$\sum_j p_j \frac{\partial x_j}{\partial p_i} + x_i = 0$$

- (3) Income changes:

$$\sum_i p_i \frac{\partial x_i}{\partial w} = 1$$

Indirect Utility Function

- Defined as

$$v(p, w) = \max_{x \in \mathbb{R}_+^n} u(x) \quad \text{s.t. } p \cdot x \leq w$$

- Value of consumer problem
- Maximum utility attainable at (p, w)

Proposition 7: Properties of $v(p, w)$

Assume: u continuous, preferences locally non-satiated

- (1) Homogeneous of degree 0: $v(\lambda p, \lambda w) = v(p, w)$
- (2) Continuous in (p, w) , $p \gg 0$, $w \geq 0$
- (3) Nonincreasing in p , strictly increasing in w
- (4) Quasi-convex in (p, w)

Proof of Proposition 7 (1–2)

(1) Homogeneity

- Scaling p, w by λ does not change budget set
- Same feasible set $\Rightarrow$ same maximum utility

(2) Continuity

- Take sequences $(p^n, w^n) \rightarrow (p, w)$
- Show $\liminf v(p^n, w^n) \geq v(p, w)$ by constructing feasible bundles
- Show $\limsup v(p^n, w^n) \leq v(p, w)$ using compactness of feasible sets
- Conclude $\lim v(p^n, w^n) = v(p, w)$

Proof of Proposition 7 (3–4)

(3) Monotonicity

- If $p' > p$, then $B(p', w) \subset B(p, w)$
- Smaller choice set $\Rightarrow v(p', w) \leq v(p, w)$
- If $w' > w$, then $B(p, w) \subset B(p, w')$
- Larger budget $\Rightarrow$ strictly higher utility

(4) Quasi-convexity

- Suppose $v(p, w) \leq v$ and $v(p', w') \leq v$
- For any $t \in [0, 1]$, let $(p_t, w_t) = t(p, w) + (1 - t)(p', w')$
- Any x feasible at (p_t, w_t) is feasible at least for one of (p, w) or (p', w')
- So $u(x) \leq v$, hence $v(p_t, w_t) \leq v$

Lagrangian Approach

- Consumer problem with differentiable $u(x)$
- Lagrangian:

$$L(x, \lambda, \mu; p, w) = u(x) + \lambda(w - p \cdot x) + \sum_i \mu_i x_i$$

- $\lambda \geq 0$: multiplier on budget
- $\mu_i \geq 0$: multipliers on $x_i \geq 0$

Kuhn–Tucker Conditions

- FOCs: $\frac{\partial u}{\partial x_k} = \lambda p_k - \mu_k$
- Constraints: $p \cdot x \leq w, x \geq 0$
- Complementary slackness:

$$\lambda(w - p \cdot x) = 0, \quad \mu_k x_k = 0$$

Proposition 8: Kuhn–Tucker

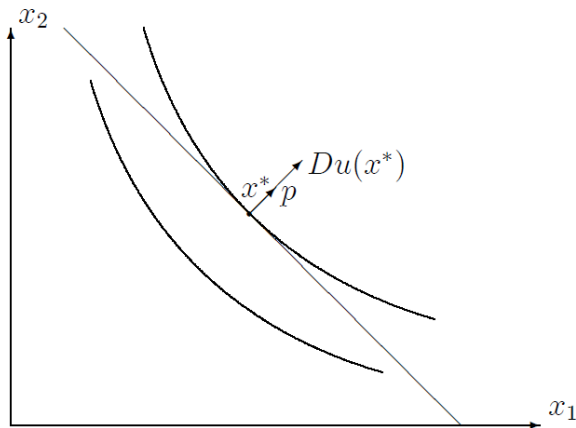
Statement

- If u differentiable, constraint qualification holds
- Then $\exists(\lambda, \mu) \geq 0$ s.t. (x, λ, μ) solves KT
- If u quasi-concave and gradient condition holds
- Then $\text{KT} \iff \text{consumer problem solution}$

Proof (concise)

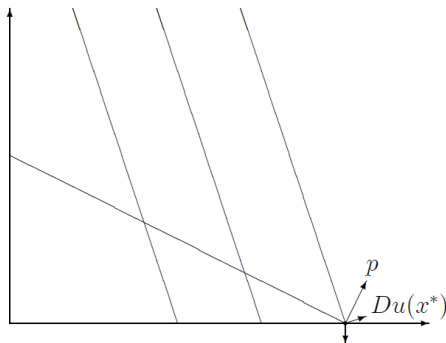
- Existence from Kuhn–Tucker theorem
- Sufficiency: quasi-concavity $\Rightarrow$ global optimality

Proposition 8: Interior Solution



- Both goods consumed in positive amounts
- $\mu_i = 0$ for all i
- $MRS = \text{price ratio}$

Proposition 8: Corner Solution



Gradient of the $-x_1 \leq 0$ constraint.

- Some goods not consumed ($x_k = 0$)
- Corresponding $\mu_k > 0$
- $\text{MRS} \neq \text{price ratio}$

Envelope Theorem

- Setup: $V(\theta) = \max_{x \in X(\theta)} f(x, \theta)$
- $x^*(\theta)$ optimal choice
- Envelope Theorem:

$$\frac{dV(\theta)}{d\theta} = \frac{\partial f(x^*(\theta), \theta)}{\partial \theta}$$

- Intuition:
 - Ignore effect of $x^*(\theta)$ changing
 - Derivative depends only on direct effect of θ
- Applications in consumer theory:
 - $\partial v(p, w)/\partial w = \lambda$
 - $\partial v(p, w)/\partial p_i = -\lambda x_i$

Proposition 9: Marginal Utility of Wealth

Statement

- If u continuous, quasi-concave, $p \gg 0$
- Unique (x, λ, μ) solves Lagrangian
- Then v differentiable at (p, w) and

$$\frac{\partial v(p, w)}{\partial w} = \lambda \geq 0$$

Proof

- Envelope theorem: derivative wrt w equals λ

Proposition 10: Roy's Identity

Statement

- If v differentiable, $\partial v / \partial w > 0$
- Then demand singleton
- Roy's Identity:

$$x_i(p, w) = - \frac{\partial v(p, w) / \partial p_i}{\partial v(p, w) / \partial w}$$

Proof

- Envelope theorem: $\frac{\partial v}{\partial p_i} = -\lambda x_i$
- Also $\frac{\partial v}{\partial w} = \lambda$
- Solve ratio $\Rightarrow$ Roy's Identity

Summary of Section 4

- Lagrangian $\Rightarrow$ KT conditions
- Interior vs corner solutions
- λ : marginal utility of wealth
- Roy's Identity links $v(p, w)$ to demand

Expenditure Minimization Problem (EMP)

- Consumer minimizes expenditure to reach utility u

$$\min_{x \in \mathbb{R}_+^n} p \cdot x \quad \text{s.t.} \quad u(x) \geq u$$

- Solution: Hicksian (compensated) demand $h(p, u)$
- Value: Expenditure function

$$e(p, u) = \min_x \{p \cdot x : u(x) \geq u\}$$

Proposition 11: Existence

Statement

- If $p \gg 0$, u continuous, and $\exists x$ with $u(x) \geq u$
- Then (EMP) has a solution

Proof (concise)

- Feasible set closed, bounded, non-empty
- Continuous $p \cdot x$ achieves minimum

Proposition 12: Properties of Hicksian Demand

- (1) Homogeneous of degree 0 in p : $h(\lambda p, u) = h(p, u)$
- (2) No excess utility: $u(h(p, u)) = u$
- (3) Convexity: if preferences convex, $h(p, u)$ convex
- Strict convexity $\Rightarrow$ unique solution

Proof ideas

- (1) Scaling prices multiplies objective by λ only
- (2) If $u(x) > u$, scale down x to cheaper bundle with $u \geq u$
- (3) Intersection of convex sets is convex

Proposition 13: Properties of Expenditure Function

- (1) Homogeneous of degree 1 in p
- (2) Continuous in (p, u)
- (3) Nondecreasing in p , strictly increasing in u
- (4) Concave in p

Proof ideas

- (1) Scale prices by λ , minimum cost scales by λ
- (3) Higher $p \Rightarrow$ higher cost; higher $u \Rightarrow$ strictly more needed
- (4) Concavity from cost minimization

Proposition 14: Shephard's Lemma

Statement

- If $h(p, u)$ singleton
- Then $e(p, u)$ differentiable in p
- And

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u)$$

Proof (concise)

- Envelope theorem: derivative wrt p_i equals optimal x_i
- So Hicksian demand recovered from expenditure function

Summary of Hicksian Demand

- EMP: minimize expenditure to reach utility u
- Hicksian demand $h(p, u)$: optimal bundle
- Expenditure function $e(p, u)$: minimum cost
- Properties:
 - h : homogeneous degree 0, convex, exact utility
 - e : homogeneous degree 1, monotone, concave
- Shephard's Lemma: $h_i = \partial e / \partial p_i$

Comparative Statics

- Parameters: prices p , wealth w
- Study how demand and expenditure respond
- Hicksian demand $h(p, u)$
- Expenditure function $e(p, u)$

Proposition 15: Law of Demand

Statement

- If $x \in h(p, u)$ and $x' \in h(p', u)$
- Then

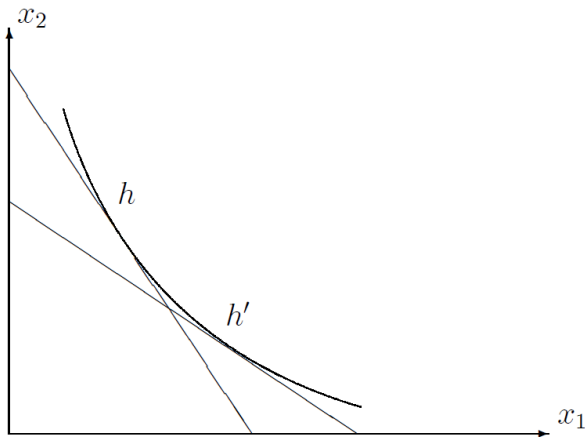
$$(p' - p) \cdot (x' - x) \leq 0$$

- Own-price Hicksian demand weakly decreases in price

Proof (concise)

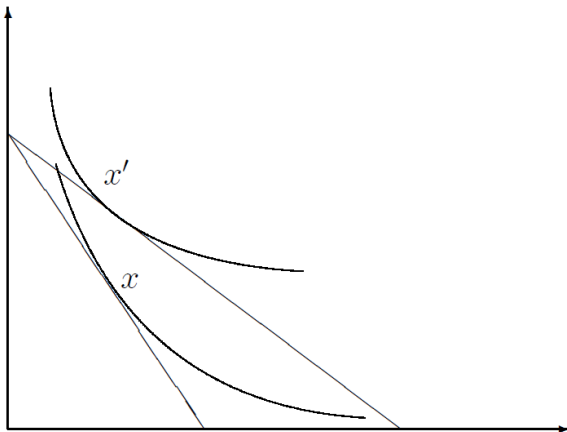
- $p' \cdot x' \leq p' \cdot x$, and $p \cdot x \leq p \cdot x'$
- Rearrange inequalities to get result

Figure: Hicksian Law of Demand



- Price increase $\Rightarrow$ move along indifference curve
- Hicksian demand decreases in own price

Marshallian vs Hicksian Demand



- Hicksian: substitution effect only
- Marshallian: substitution + wealth effect
- Marshallian demand can slope upward

Proposition 16: Substitution Matrix

Definition

$$D_p h(p, u) = \begin{pmatrix} \partial h_1 / \partial p_1 & \cdots & \partial h_1 / \partial p_n \\ \vdots & \ddots & \vdots \\ \partial h_n / \partial p_1 & \cdots & \partial h_n / \partial p_n \end{pmatrix}$$

Statement

- If $h(p, u)$ differentiable
- Then $D_p h(p, u)$ symmetric
- And negative semi-definite

Proof Sketch of Proposition 16

- By Shephard's Lemma: $h_i = \partial e / \partial p_i$
- So $\partial h_i / \partial p_j = \partial^2 e / \partial p_i \partial p_j$
- (1) Symmetry: Young's theorem $\Rightarrow f_{xy} = f_{yx}$
- (2) Negative semi-definite: $e(p, u)$ concave in p
- Diagonal ≤ 0 : demand weakly decreasing in own price

Summary of Section 6

- Hicksian law of demand: demand curves slope downward
- Marshallian demand can slope upward (Giffen goods)
- Substitution matrix symmetric and negative semi-definite
- Cross-price effects symmetric across goods

Marshallian vs Hicksian Demand

- Marshallian: $x(p, w)$ (utility max, budget constraint)
- Hicksian: $h(p, u)$ (cost min, target utility u)
- Connected via indirect utility $v(p, w)$ and expenditure function $e(p, u)$

Proposition 17: Duality Relations

Statement

- For all $p \gg 0$, $w \geq 0$:

$$x(p, w) = h(p, v(p, w)), \quad e(p, v(p, w)) = w$$

- For all $p \gg 0$, $u \geq u(0)$:

$$h(p, u) = x(p, e(p, u)), \quad v(p, e(p, u)) = u$$

Proof idea

- By local non-satiation: budget binds, $p \cdot x = w$
- If $u = v(p, w)$, then $e(p, u) = w$
- Similarly, if $w = e(p, u)$, then $v(p, w) = u$

Proof of Proposition 17

To show:

$$x(p, w) = h(p, v(p, w)), \quad e(p, v(p, w)) = w$$

Proof sketch

- Let $x \in x(p, w) \Rightarrow u(x) = v(p, w), p \cdot x \leq w, e(p, v(p, w)) \leq w$
- Local non-satiation $\Rightarrow p \cdot x' < w$ implies $u(x') \leq v(p, w)$
- Thus $e(p, v(p, w)) \geq w$
- Hence $e(p, v(p, w)) = w$ and $x = h(p, v(p, w))$

Symmetry

- If $x \in h(p, u) \Rightarrow u(x) = u, p \cdot x = e(p, u)$
- Then $v(p, e(p, u)) = u$, so $h(p, u) = x(p, e(p, u))$

Proposition 18: Slutsky Equation

Statement

$$\frac{\partial x_i(p, w)}{\partial p_j} = \frac{\partial h_i(p, u)}{\partial p_j} - \frac{\partial x_i(p, w)}{\partial w} x_j(p, w)$$

- $u = v(p, w)$
- LHS: total effect of price change
- RHS: substitution effect – wealth effect

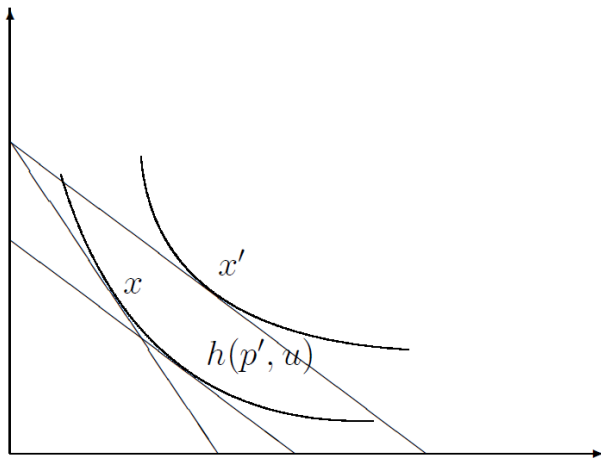
Proof idea

- Start: $h_i(p, u) = x_i(p, e(p, u))$
- Differentiate wrt p_j
- Apply Shephard's lemma $\partial e / \partial p_j = h_j(p, u)$
- Rearrange $\Rightarrow$ Slutsky equation

Interpretation of Slutsky Equation

- Total price effect = Substitution effect + Wealth effect
- Substitution effect: move along indifference curve
- Wealth effect: change in demand due to real income change
- Normal vs inferior goods: direction of wealth effect

Figure: Substitution and Wealth Effects



- Price drop: $p_1 \rightarrow p'_1$
- Demand moves from $x \rightarrow h(p', u) \rightarrow x'$
- Decomposition: substitution effect + wealth effect

Definitions

- **Normal good:** $x_i(p, w)$ increases in w
- **Inferior good:** $x_i(p, w)$ decreases in w
- **Regular good:** $x_i(p, w)$ decreases in p_i
- **Giffen good:** $x_i(p, w)$ increases in p_i
- **Substitute:** $h_i(p, u)$ increases in p_j
- **Complement:** $h_i(p, u)$ decreases in p_j
- **Gross substitute:** $x_i(p, w)$ increases in p_j
- **Gross complement:** $x_i(p, w)$ decreases in p_j

Summary of Section 7

- Marshallian and Hicksian demand linked by duality
- Slutsky equation: price effect = substitution + wealth
- Substitution matrix symmetric, negative semi-definite
- Definitions: normal, inferior, Giffen, substitutes, complements

Welfare and Price Changes

- Question: how much better/worse off after price change $p \rightarrow p'$
- Natural measure: $\Delta v = v(p', w) - v(p, w)$
- Problem: depends on utility representation (in utils)
- Solution: measure in monetary terms
- Tools: compensating variation (CV), equivalent variation (EV)

Compensating Variation (CV)

- Reference utility: before price change
- $u = v(p, w)$
- Definition:

$$CV = e(p, u) - e(p', u) = w - e(p', u)$$

- Interpretation: how much wealth to take away after price change
- Keeps consumer on old indifference curve

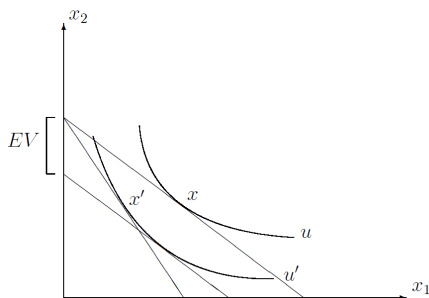
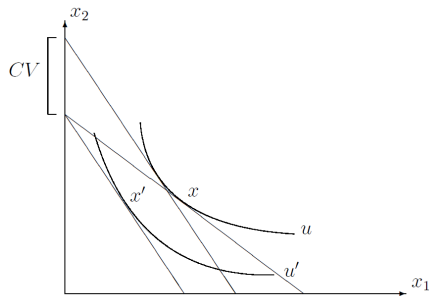
Equivalent Variation (EV)

- Reference utility: after price change
- $u' = v(p', w)$
- Definition:

$$EV = e(p, u') - e(p', u') = e(p, u') - w$$

- Interpretation: how much extra wealth before price change
- Makes consumer as well off as after price change

Figures: CV and EV



- CV: adjust wealth after change to keep u
- EV: adjust wealth before change to reach u'

Relation to Hicksian Demand

- CV and EV computed from Hicksian demand
- Compensating Variation:

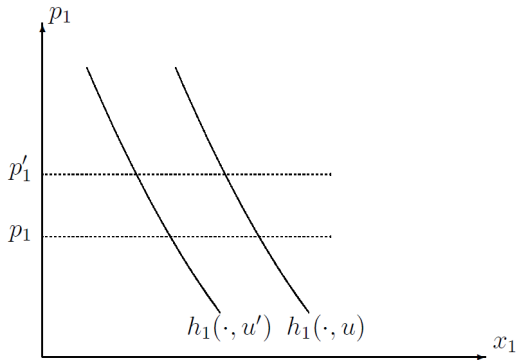
$$CV = \int_{p'_i}^{p_i} h_i(p, u) dp_i$$

- Equivalent Variation:

$$EV = \int_{p'_i}^{p_i} h_i(p, u') dp_i$$

- Areas under Hicksian demand curves

Figure: Hicksian Demand and Welfare



- $h_1(p, u)$ and $h_1(p, u')$
- $CV = \text{area left of } h_1(p, u)$
- $EV = \text{area left of } h_1(p, u')$

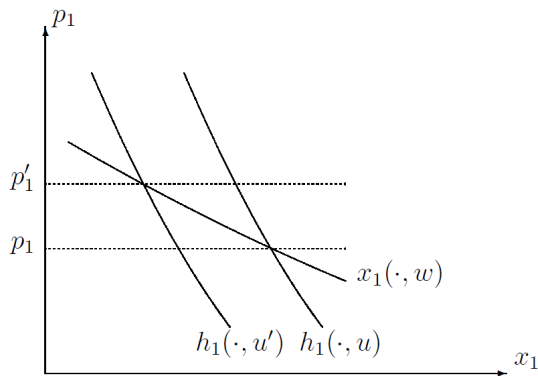
Consumer Surplus

- Define: area under Marshallian demand

$$CS = \int_{p'_i}^{p_i} x_i(p, w) dp_i$$

- Approximates welfare change
- Lies between CV and EV (if good normal/inferior)
- Widely used in applied work

Figure: Marshallian Surplus



- Marshallian demand curve
- CS between CV and EV

Summary of Section 8

- Welfare change from prices: use CV , EV , or CS
- CV : wealth adjustment after change
- EV : wealth adjustment before change
- Hicksian demand used to compute CV , EV
- CS : approximation, lies between CV and EV